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ABSTRACT

We report the molecular beam epitaxial growth and characterization of wurtzite InScN thin films. We find that wurtzite InScN can be grown on GaN substrates under metal-rich conditions, resulting in smooth surfaces with <1 nm roughness and a two-dimensional growth mode. We further find that InScN can be stabilized in the wurtzite crystalline phase for Sc content up to at least 10.4%. We measure and report the lattice parameters of wurtzite InScN as a function of Sc content and find that between ∼5% and 8% Sc mole fraction InScN has a similar a lattice constant to InN. The c/a ratio deviates from the ideal tetrahedral value of 1.633 as Sc content increases. The optical absorption edge of wurtzite InScN monotonically increases over a Sc content range of 0%–10.4% from 0.8 to 1.05 eV, accompanied by a shift in the peak index of refraction and extinction coefficient. The E_2^{high} Raman peak of InScN shows negligible shift with increasing Sc content compared to a gradual increase in the A_1 (LO) peak position. These fundamental observations can facilitate the future design of InN-based devices.

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I. INTRODUCTION

The transition metal scandium (Sc) has garnered significant attention in the field of electronics and photonics due to its ability to enhance the piezoelectric coefficient, non-linear optical properties, and ferroelectric nature in conventional III-N binary semiconductor materials such as AlN and GaN by alloying.[1–4] The ideal substitution of Sc into the III-N binaries is in part due to the favorable +3 oxidation state of Sc.[5] These properties have enabled a wide range of applications for Sc alloyed nitrides, such as bulk-acoustic wave (BAW) resonators, high-electron mobility transistors (HEMTs), and distributed Bragg reflectors.[6–8] High-performing, functional devices using AlScN have become possible due to extensive research in understanding the growth mechanisms and the subsequent measurement of material properties in high quality thin films.

For example, critical work has been done in plasma-enhanced molecular beam epitaxy (MBE) on the role of Sc content in AlScN in the transition between the rock salt and wurtzite crystalline phases and the need for nitrogen-rich conditions to prevent intermetallic formation in AlScN.[9] The use of indium as a surfactant in MBE enables improved surface morphology for AlScN.[10] GaScN has not been used as extensively in devices yet, though its growth has received attention. Metal-rich conditions were found to allow the 2D growth of wurtzite phase GaScN without secondary phase formation.[11] Compared to prior studies of AlScN and GaScN, there have been no experimental reports of InScN grown by MBE; however, while this manuscript was under revision, a recent study reporting the sputter deposition of InScN was published.[12]

InN offers unique properties not seen in the likes of GaN and AlN, such as its small, direct bandgap of 0.7 eV and exceptionally low effective mass[13,14] with reports as low as $0.042m_e$. In addition,

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p-type doping inferred from photoconductivity measurements has been reported for InN via MBE, which had previously been shown to be quite difficult.[15] Use of p-doping for compensation has enabled the observation of 2D electron gases in InN.[16] The optical properties of InN have been studied for infrared photodetectors[17] and solar cells[18] by integrating it with other materials.

But, the two main In containing III-nitride ternaries of InAlN and InGaN are quite difficult to integrate with InN. This is because to achieve low-strain heterostructures, InAlN and InGaN need to be grown with high In compositions, at which there is significant phase segregation due to the high miscibility gap between InN and AlN or GaN.[19–21] Thus, to enable a new generation of InN-based devices, a new ternary should be explored that possesses a stable wurtzite phase at high In-composition without phase segregation.

In this work, we first establish the molecular beam epitaxial growth conditions to deposit InScN thin films. Using these films, we determine the critical Sc mole fraction window for the phase transition from the wurtzite phase to a mixed wurtzite/rock salt phase. By studying the surface morphology and the reflection high-energy electron diffraction (RHEED) pattern, the viability of metal-rich conditions to promote a 2D growth mode without the formation of intermetallics or other secondary phases is inspected. In addition, the effects of Sc content on the lattice constants and crystal quality for the pure wurtzite phase samples of InScN are determined. The optical properties such as the index of refraction, extinction coefficient, and absorption edge of the wurtzite InScN samples are also measured. Finally, the Raman spectrum is reported to gain insight on the effect that Sc has on the phonon behavior in wurtzite InScN. This work provides not only the fundamental guide to the MBE growth of InScN for future studies but also an index of the essential material properties that would allow for the design of novel electronics and photonics.

II. EXPERIMENTAL

All InScN films in this work were grown under In-rich conditions via plasma-assisted MBE. As stated earlier, metal-rich conditions are not viable for the growth of AlScN due to the formation of intermetallics.[9] In the case of AlScN, Al readily displaces Sc during the growth, which then results in an accumulation of Al and Sc on the surface that reacts to form Al_3Sc. For InScN, the process is quite different. The formation energies for InN, In, rock salt ScN, and Sc are −140.7, −76.4, −213.1, and −43.3 kJ/mol, respectively. For the reaction ScN + In ⇒ InN + Sc, the net change in Gibbs free energy of formation is $\Delta G_f = 105.5$ kJ/mol.[9,22] This implies that Sc will preferentially displace In during the growth of InScN. Thus, growing under In-rich conditions will result in excess In on the surface with Sc incorporating into the film. The allowed metal-rich growth condition promotes a 2D step-flow growth mode that is not possible in nitrogen-rich conditions often used for AlScN.[23,24]

In this study, the InScN samples were grown by plasma-assisted MBE using a Veeco Gen10 MBE reactor on *c*-plane, Ga-polar bulk n-type GaN substrates. During the growth, the surfaces of the films were monitored *in situ* using KSA Instruments (RHEED) paired with a Staib electron gun operated at 14.5 kV, 1.45 A. Nitrogen gas of 99.999 95% purity used in the growth was provided using an RF plasma source with a flow rate of 1.05 SCCM and 200 W RF power. Scandium (99.999%), indium (99.999 99%), silicon (99.9999%), and gallium (99.999 99%) were supplied using separate effusion cells.

Before the growth of InScN, a 40 nm n+GaN buffer layer was grown under metal-rich conditions at a substrate temperature measured by a thermocouple $T_C = 700\,^{\circ}C$ following a Ga polish of the underlying substrate at 750 °C. Excess Ga was thermally desorbed at 750 °C before the subsequent growth of 160 nm of InScN at 400 °C as confirmed by monitoring RHEED. The 160 nm thickness was chosen to ensure that the InN film with 0% Sc is fully relaxed, as the maximum thickness for pseudomorphic InN on GaN is 1 monolayer (ML) due to the ∼ 11% lattice mismatch.[25] Since InN decomposes at high temperatures,[26] the InScN films were all grown under In-rich conditions at a relatively low temperature of 400 °C. Furthermore, the Sc cell temperature was varied between samples to achieve Sc mole fractions up to 15.1%. The grown heterostructure is seen in Fig. 1(a).

The surface morphology was measured *ex situ* with an Asylum Research Cypher ES atomic force microscope (AFM). The

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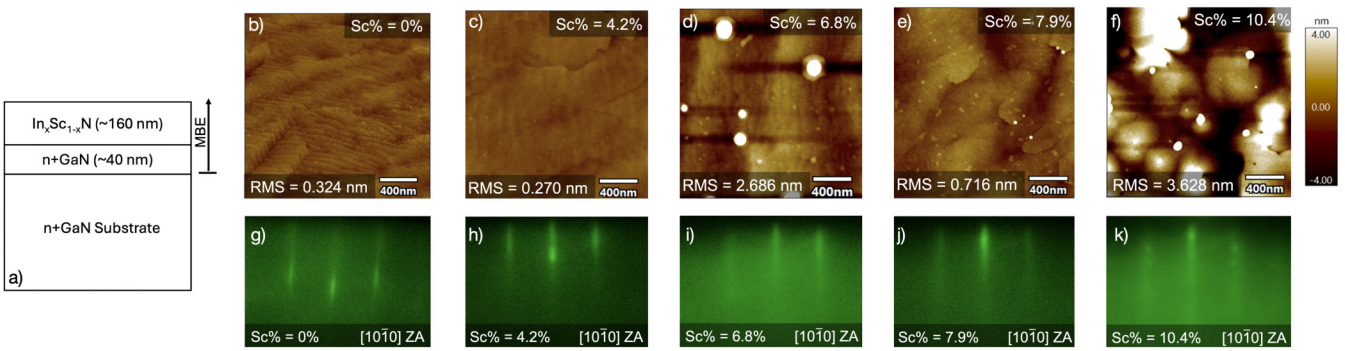


FIG. 1. (a) As grown material stack for InScN samples. $2 \times 2\,\mu m^2$ AFM micrographs for InScN films with Sc contents of (b) 0%, (c) 4.2%, (d) 6.8%, (e) 7.9%, and (f) 10.4%. RHEED patterns collected along the [10$\bar{1}$0] zone axis at the end of the growths for the (g) 0%, (h) 4.2%, (i) 6.8%, (j) 7.9%, and (k) 10.4% Sc mole fraction samples.

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crystal structure and phase purity of the InScN films were determined by x-ray diffraction (XRD). In addition, XRD was used to check for peaks corresponding to rock salt InScN and intermetallics. Furthermore, reciprocal space mapping (RSM) was performed to extract lattice constants and to detect twinning rock salt phase formations. XRD and RSM scans were collected with a PANalytical Empyrean system with Cu Kα1 radiation. The chemical composition of the InScN films was measured by x-ray photoelectron spectroscopy (XPS) using a ThermoFisher Scientific Nexsa G2 surface analysis system. The chemical compositions measured by XPS were calibrated with Rutherford backscattering spectrometry (RBS) performed at Eurofins EAG Laboratories with an uncertainty of 2% for Sc.

The optical absorption measurements were performed with a Cary 5000 UV–Vis–NIR spectrophotometer, where the absorbance A was converted to the absorption coefficient α by the relationship $\alpha = 2.303A/L$, where L is the thickness of the film.[27] The ellipsometry data used to find the index of refraction and extinction coefficient of the samples were collected with a Woollam RC2 spectroscopic ellipsometer. Raman measurements were performed using a Witec Alpha300R system equipped with a 488 nm laser. A spectrometer with 1800 gratings/mm was used to collect the scattered signal. The tool was operated in the $z(xx)\bar{z} + z(xy)\bar{z}$ scattering configuration.

III. RESULTS AND DISCUSSION

Figure 1(b) shows a $2 \times 2\,\mu m^2$ AFM micrograph for the 0% Sc film, which corresponds to InN. Since the film was grown under In-rich conditions to induce a step-flow growth mode, the surface exhibits atomic step features with an overall roughness of 0.324 nm. The presence of atomic steps is true for Sc concentrations of up to 6.8% Sc as seen in Figs. 1(c) and 1(d). For Figs. 1(e) and 1(f), the atomic steps are suppressed. The surface roughness values for the 4.2%, 6.8%, 7.9%, and 10.4% samples are 0.270, 2.686, 0.716, and 3.628 nm, respectively. Excluding the 6.8% Sc mole fraction sample, the overall surface roughness is stays below 1 nm for Sc contents below 10.4%. The overall smoothness is due to the excess In on the surface as a surfactant during the In-rich growth, similar to Ga acting as a surfactant during the growth of Ga-rich GaScN.[11] For Fig. 1(f), we see the onset of significant roughness at a value of 3.628 nm. Similar change in surface morphology and surface roughness in Sc-containing ternary wurtzite alloys are often attributed to an introduction of an additional structural phase.

Figures 1(g)–1(k) show the RHEED patterns collected from the [10$\bar{1}$0] zone axis InScN samples up to 10.4% Sc mole fraction. For all samples, a streaky diffraction pattern is seen indicating step-flow growth and continuity of the wurtzite crystal structure.[28] The streaky diffraction pattern is unlike the spotty pattern for N-rich growth of AlScN where it indicates a 3D island growth mode.[9,28] Close inspection of Fig. 1(k) shows that there is a faint spot-modulated streak diffraction pattern present for the highest Sc content InScN sample. The ordering of the spots is akin to the pattern seen in the growth of twinned rock salt ScN.[29,30] The twinning is a consequence of geometrical constraints needed for growing a threefold symmetric 111-oriented cubic crystal on a sixfold symmetric 002-oriented wurtzite crystal.[29,30] Therefore, the RHEED pattern indicates a possible onset of a mixed wurtzite/rock salt phase for the 10.4% Sc mole fraction sample, potentially explaining the changes in surface morphology observed in Fig. 1(f). Further confirmation of secondary phase formation is discussed with the XRD data later in this work.

Figure 2(a) shows the symmetric ω–2θ XRD scans for all grown films in the study. For all samples with Sc mole fraction below 15.1%, there are only two observable peaks. The peak corresponding to the 002 plane of wurtzite GaN is at ~17.28°, and the other peak corresponds to wurtzite InScN. The InScN 002 peak shifts from 15.617° to 15.798° in the range from 0% to 10.4%. However, once the Sc mole fraction reaches 15.1%, the formation of a secondary phase along with the 002 wurtzite peak becomes apparent at a peak location of 16.84°. We hypothesize that a secondary rock salt phase of InScN is present for the 15.1% Sc content InScN sample. The lattice parameter expected for the rock salt phase of ScN is 4.602 Å; XRD data discussed later will test this

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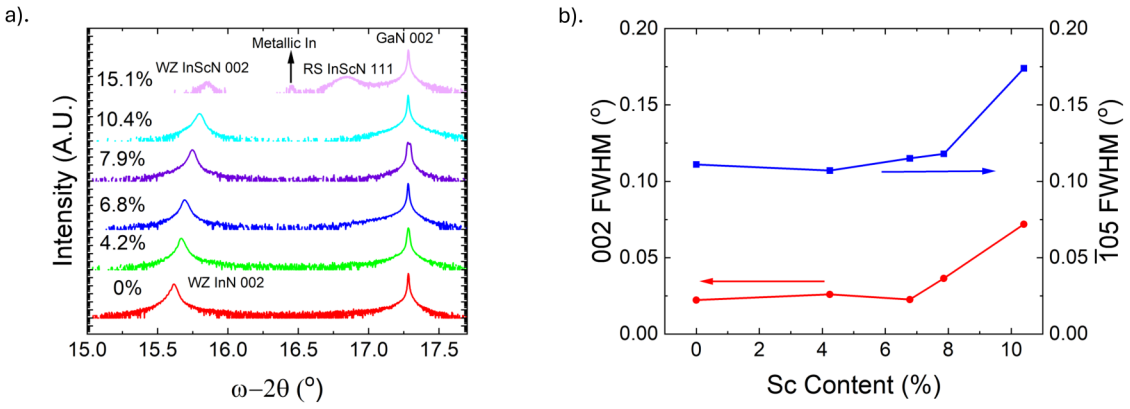


FIG. 2. (a) Symmetric ω–2θ XRD of the 002 peak for both GaN and InScN. The 15.1% Sc mole fraction sample also has an additional 111 rock salt and metallic In peak. (b) Full-width at half maximum (FWHM) for the 002 and −105 rocking curve (RC) for pure wurtzite InScN phase samples (<15.1% Sc).

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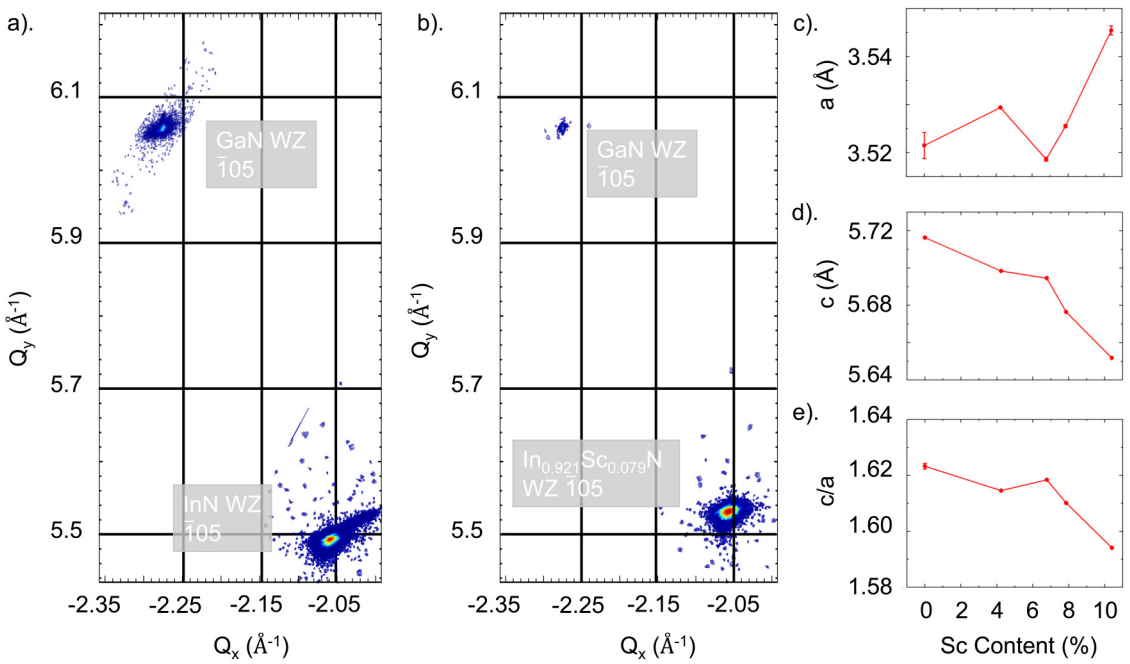


FIG. 3. (a) RSM showing the −105 peak for InN and GaN. (b) RSM showing the −105 peak for $In_{0.921}Sc_{0.079}N$ and GaN. (c) Plot for the extracted in-plane lattice constant *a* for all pure wurtzite samples. (d) Plot for the extracted out-of-plane lattice constant *c* for all pure wurtzite samples. (e) Plot for the *c*/*a* ratio for all pure wurtzite samples.

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hypothesis. The Sc composition between 10.4% and 15.1% observed here for the phase transition from wurtzite to rock salt is quite different from the ∼40% Sc mole fraction theoretical prediction.[31] An additional diffraction peak is seen at ∼ 16.5° for the 15.1% Sc sample. The peak likely corresponds to metallic In as seen in XRD patterns for In-rich grown InGaN and InN.[32,33] Furthermore, the same metallic In peak is seen for the diffraction pattern of pure In powder.[34] However, the metallic In peak is not present for the other samples as the surfaces were cleaned with HCl. Unless noted, XRD scans were carried out post-HCl treatment for In droplet removal. The 15.1% Sc content sample could not be cleaned in HCl as we observe film delamination.

Figure 2(b) shows the full-width at half maximum (FWHM) of the 002 InScN ω rocking curve peak for all of the pure wurtzite InScN samples. The 002 rocking curve peak FWHM is primarily affected by pure screw dislocations with a line direction of [001] since these dislocations affect (hkl) planes for which l is nonzero.[35,36] It is also possible for mixed dislocations to increase the FWHM.[35,36] From Fig. 2(b), the FWHM is relatively constant between 0% and 6.8% Sc mole fraction at a value between 0.02° and 0.04°. But, between 6.8% and 10.4% Sc mole fractions, the FWHM increases from ∼0.02° up to ∼0.07°, implying that higher levels of Sc introduce more screw or mixed dislocations. Figure 2(b) also shows a similar trend for the FWHM of the $\bar{1}05$ rocking curve. Since threading dislocations only affect planes with either non-zero h or k, the FWHM increases with the density of edge and mixed threading dislocations.[36] The FWHM stays relatively constant at a value between 0.10° and 0.12° from 0% to 6.8% Sc mole fraction. Past a Sc mole fraction of 6.8%, the FWHM goes from 0.12° to 0.17°, indicating an increase in the edge or mixed dislocation density.

Figure 3(a) displays the RSM scans containing the $\bar{1}05$ peaks for both wurtzite InN and wurtzite GaN. Figure 3(b) shows the $\bar{1}05$ peaks for wurtzite $In_{0.921}Sc_{0.079}N$ and GaN. A significant shift is seen in the InScN peak position compared to InN in the Q_y direction as opposed to Q_x. This indicates a larger change in the out-of-plane lattice constant than in the in-plane lattice constant with increasing Sc incorporation.

Figure 3(c) shows the in-plane lattice constant *a* as extracted from RSM for all of the single-phase wurtzite samples. The in-plane lattice constant calculated from the RSM for InN is 3.521 Å, which falls within the range of 3.483–3.585 Å for the reported values for InN.[37–39] Interestingly, it would appear from Fig. 3(c) that InScN has a similar *a* lattice constant to InN with Sc alloy composition between ∼5% and 8% Sc mole fractions. The behavior of a ternary sharing a similar in-plane lattice constant to its parent binary is not unheard of as B-alloyed AlN exhibits the same feature.[40]

Overall, the change of the *a* lattice parameter is relatively small, with the total range being from 3.519 to 3.545 Å. In Fig. 3(d), the extracted out-of-plane lattice constant *c* varies significantly more with Sc composition than the *a* lattice parameter. The total range for the *c* lattice parameter is 5.716–5.652 Å. Similar to the *a* lattice parameter, the *c* lattice parameter of the InN sample lies within the

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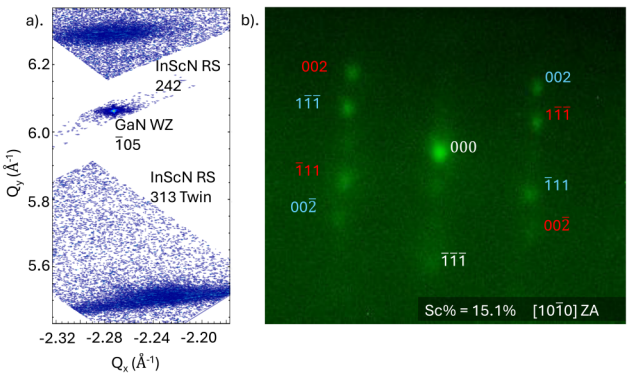


FIG. 4. (a) RSM showing the −105 peak for GaN along with the 242 and 313 twinning peaks for the rock salt InScN phase of the 10.6% Sc mole fraction sample. (b) RHEED pattern 14 min into the growth for the 10.6% Sc mole fraction sample showing the onset of the twinned rock salt phase.

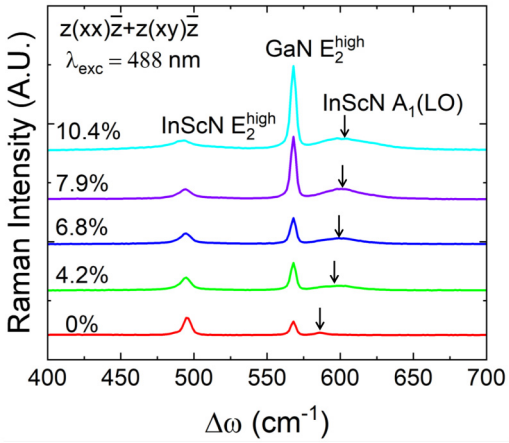


FIG. 5. Raman spectrum for all pure wurtzite samples. The data were collected in z(xx)$\bar{z}$ + z(xy)$\bar{z}$ scattering configuration with a 488 nm wavelength excitation.

reported value range of 5.54–5.8 Å for InN thin films.[37–39] The change in the a lattice constant is non-monotonic with Sc content, similar to the case of the c lattice parameter for AlScN[41,42] due to the interaction between the average bond length and tetrahedral distortion.[43] In the case of GaScN, the lattice expansion with increasing Sc content is anisotropic, indicating levels of tetrahedral distortion.[44]

Figure 3(e) shows a decrease in the c/a ratio for the InScN samples starting from 1.623 for InN to 1.594 for the 10.4% Sc mole fraction sample. Since the c/a ratio deviates further from the ideal tetrahedral value of 1.633 with increasing Sc content, the distance between centroids of the cation and anion charges will deviate from zero, increasing the spontaneous polarization.[45] Based on the c/a ratio alone, the spontaneous polarization of InScN is expected to increase with Sc content. Experimentally, a direct measurement of the spontaneous polarization value requires switching of the polarization vector. Future work will be conducted to analyze the spontaneous polarization values.

Figure 4(a) shows the RSM of the 15.1% Sc mole fraction. One can observe the clear −105 GaN wurtzite peak at $(Q_x, Q_y) = (-2.27\ \text{Å}^{-1}, 6.06\ \text{Å}^{-1})$ in addition to two additional peaks surrounding it. From the first observation, the two additional peaks are near the expected positions for the rock salt ScN 242 peak and twinned rock salt ScN 313 peak.[29,30] Upon lattice parameter calculations assuming rock salt phase, the extracted values are 4.602 and 4.608 Å for the 242 and 313 peaks, respectively. Furthermore, the lattice parameter extracted from the rock salt 111 peak in Fig. 2(a) is also equal to 4.602 Å. The agreement of lattice parameters from both the 111 peak in XRD and the two twinned asymmetric peaks in RSM confirms that the secondary phase is in fact twinned rock salt InScN, and their deviation from 4.505 Å for ScN is likely due to In incorporation.[29,30]

Figure 4(b) shows the RHEED pattern for the 15.1% Sc mole fraction sample 14 min into the growth. The two sets of peaks corresponding to the two twinned domains of rock salt InScN are visible and labeled, similar to the growths of rock salt ScN.[29] It should be noted that these domains are present at the start of the growth as verified by RHEED. Thus, Figs. 2 and 4 in conjunction justify the secondary phase of the 15.1% Sc mole fraction sample to be a twinned rock salt phase and not an intermetallic.

Figure 5 shows the collected room-temperature Raman spectra for the pure wurtzite phase InScN samples. The observed peaks are E_2^{high} for InScN and GaN as well as A_1(LO) for InScN. For the InN sample, the peaks are located at 494, 568, and 586 cm^{-1}. For GaN, from previous experimental work, the room-temperature peak for E_2^{high} was reported[46] as 567.4 cm^{-1}. Hence, the 568 cm^{-1} most likely corresponds to E_2^{high} for GaN. For InN, the reported values[47,48] for E_2^{high} phonon mode range from 488 to 495 cm^{-1}. The reported values[47,48] for A_1(LO) peak of InN range from 580 to 590 cm^{-1}. Thus, the 494 and 586 cm^{-1} peaks in the InN scan correspond to E_2^{high} and the A_1(LO) for InN, respectively. Interestingly, E_2^{high} for InScN is relatively constant at all Sc mole fractions, while the A_1(LO) InScN peak monotonically increases with Sc content. Numerically, A_1(LO) peak position for InScN goes from 586 to 597 cm^{-1}.

The Tauc plots for all pure wurtzite phase samples are shown in Fig. 6(a) along with the best fitted lines to extract the absorption edge values. For the InN sample, the absorption edge is around 0.8 eV, higher than the room-temperature bandgap of ∼0.70 eV reported in the literature.[13,14,49] The shift is due to the high background electron concentration filling of states near the bottom of the conduction band (Moss–Burstein effect).[50–52] Because the samples were grown on n+GaN, the exact number of carriers in the InScN films can not be precisely determined from Hall effect measurements. Future studies will be conducted on the carrier concentrations of InScN with the samples grown on semi-insulating GaN substrates. Regardless, the trend of the absorption edge with the Sc mole fraction should still be proportional to the changes in the optical bandgap assuming a constant background shift from the Moss–Burstein effect. Since ScN has an indirect bandgap of ∼0.92 eV[53] and a direct bandgap of ∼2.05 eV,[53,54] and InN has an ideal direct bandgap of ∼0.70 eV, the absorption edge energy of InScN is

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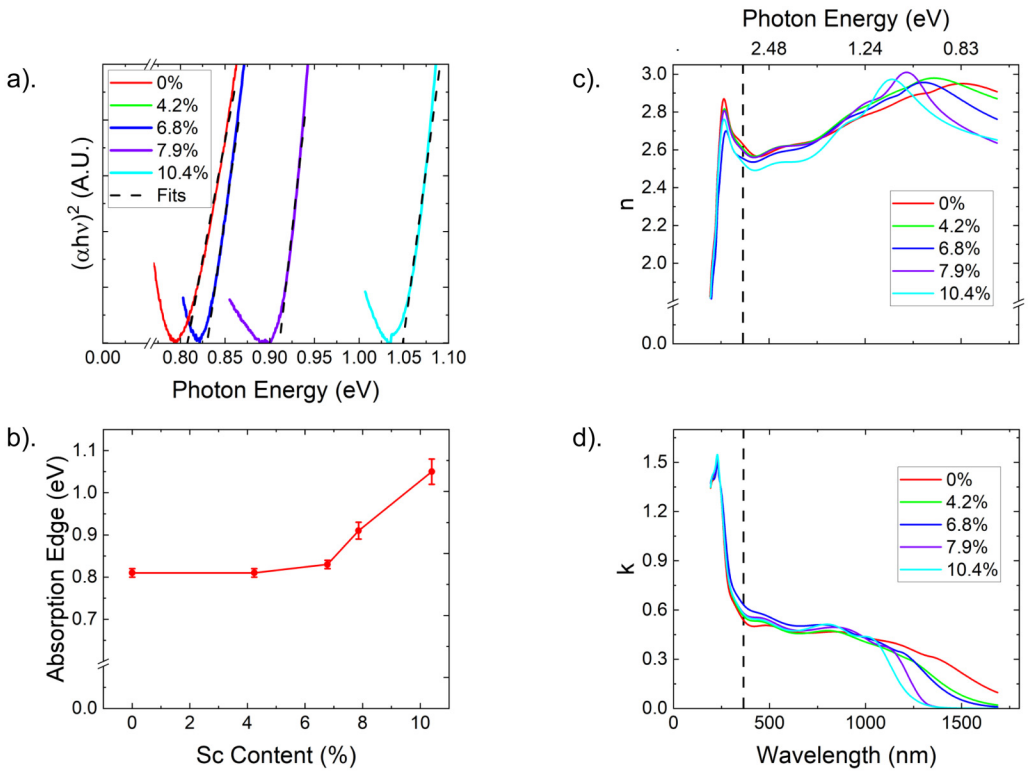


FIG. 6. (a) Tauc plot for all pure wurtzite phase samples with fitted lines to extract the absorption edge. (b) Plot of the extracted absorption edge as a function of the Sc mole fraction. (c) Index of refraction as a function of wavelength for all pure wurtzite phase samples. (d) Extinction coefficient as a function of wavelength for all pure wurtzite phase samples.

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expected to increase with Sc content. Figure 6(b) does in fact show this behavior with the absorption edge at 0.8 eV for 0% Sc InN and going to ∼1.05 eV for the 10.4% Sc mole fraction InScN sample.[30]

In Figs. 6(c) and 6(d), the index of refraction and extinction coefficient extracted from ellipsometry are plotted. The peak values for the index of refraction range from ∼2.95 to ∼3.01. Overall, the index of refraction is relatively constant with the Sc mole fraction. The biggest change is in the wavelength at which the index of refraction peaks. The wavelength range is from ∼1137 nm (1.09 eV) for the 10.4% Sc mole fraction to ∼1500 (0.83 eV) for InN. The wavelength at which the index of refraction peaks for all samples coincides with the absorption edge of the material, as seen in other studies of InN.[55] The InN peak index of refraction value of ∼2.95 agrees well with a previous report of between 2.8 and 3.0.[55] The extinction coefficients rise sharply near the absorption edge of each sample, as expected, since the extinction coefficient is directly related to the absorption. The value remains relatively constant for all samples around ∼0.45 for the visible light wavelength range.

IV. CONCLUSION

In summary, we have demonstrated the successful growth of wurtzite InScN with a Sc mole fraction of up to 10.4%. Unlike AlScN, InScN was shown to be grown in a metal-rich regime to help promote 2D step-flow growth and smooth surfaces (<1 nm roughness) for most samples. The transition point between pure wurtzite and a rock salt/wurtzite mixed phase is between a Sc mole fraction of 10.4% and 15.1%. This rock salt secondary phase is confirmed by both XRD and RSM. The a lattice parameter of the pure wurtzite phase InScN samples is similar to that of InN between ∼5% and 8% Sc mole fraction. In addition, the c/a ratio decreases from the ideal tetrahedral value of 1.633 with increasing Sc content, indicating a potential increase in the spontaneous polarization. The tunable absorption edge of InScN has been successfully measured with a tuning range of 0.8–1.05 eV. Furthermore, the index of refraction has been measured to show a relatively constant value of between 2.95 and 3.01, with the peak positions shifting with the absorption edge. Likewise, a sharp increase in the extinction coefficient is found to coincide with the shifting absorption edge. Finally, Raman spectroscopy has shown the effect of Sc on the phonon modes, with A_1 (LO) mode shifting more than E_2^{high} due to the former's polar nature. The total shift of the A_1 (LO) mode is 11 cm^{-1} for 10.4% Sc in InScN relative to InN. The detailed growth and characterization of wurtzite InScN presented here helps give the starting foundation to the development and optimization for a new-generation of InN-based devices such as HEMTs and IR photonics.

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SUPPLEMENTARY MATERIAL

See the supplementary material for the skew scan (ϕ) of the rock salt phase InScN 242 peak and the wurtzite phase InScN −105 peak for a 12.7% Sc mole fraction InScN sample, ellipsometry fit and model used to obtain the index of refraction (n) and the extinction coefficient (k), and Rutherford backscattering spectrometry showing the In, Sc, and N atomic concentration as a function of depth for the 10.4% Sc mole fraction sample.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

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DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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